

KW007 Radar Sensor GUI User Guide

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1. Introduction

The KW007 GUI application is a configuration and monitoring tool for the KW007 radar sensor module. This software allows you to:

- Configure sensor detection parameters
- Monitor real-time detection status

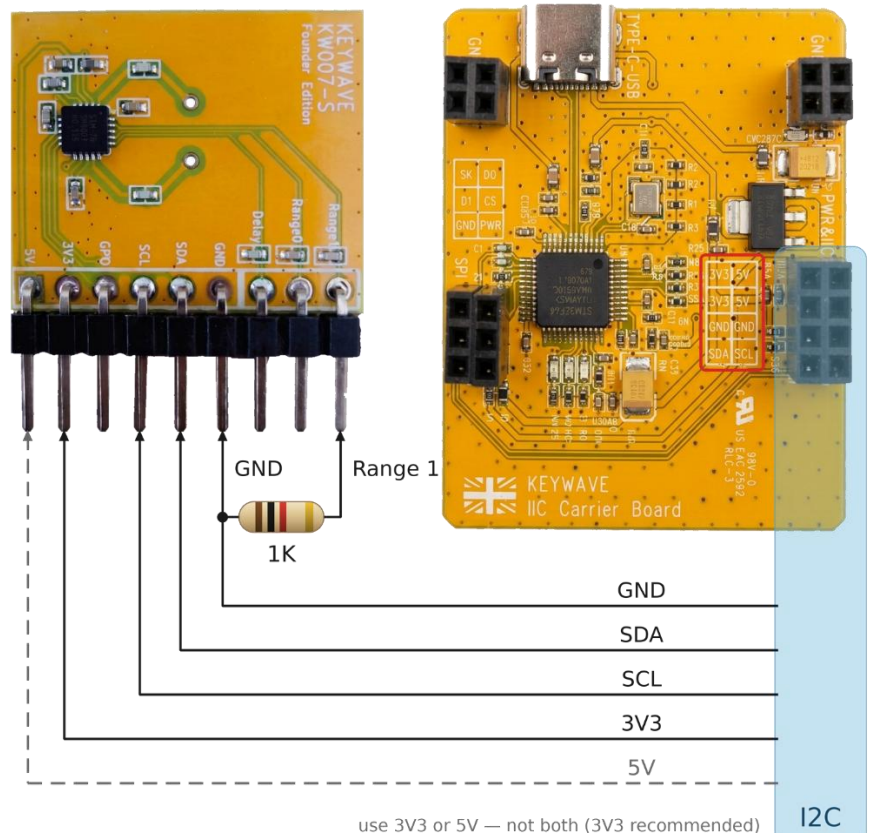
2. System Requirements

- **Operating System:** Windows 10 or later
- **Hardware:** FT232H or CH341 USB-to-I²C adapter
- **Driver:** FTDI D2XX driver (FT232H) or WCH CH341PAR driver (CH341) installed
- **Connection:** KW007 module connected via I²C

3. Getting Started

3.1 Connecting the Hardware

1. Connect the USB-to-I²C adapter to your computer
2. Connect the KW007 module to the I²C pins:
 - VCC (Power) — **(3.3 V recommended)**
 - SDA (Data)
 - SCL (Clock)
 - GND (Ground)
 - Range1 (Ground)



KEYWAVE

3.2 Hardware Note

When using KW007 in **I²C mode**, the module's Range1 pin must be connected to GND through a resistor of approximately 1 kΩ.

3.3 Power Supply Note

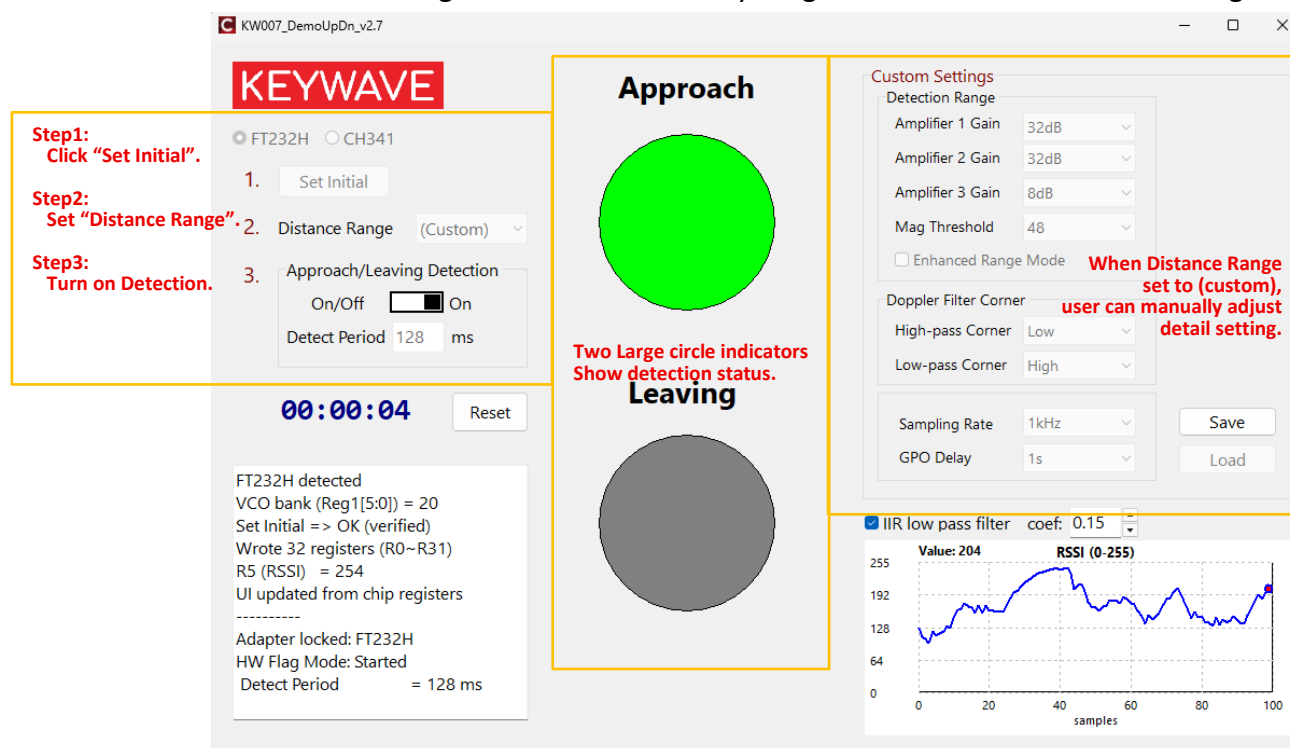
Use either the 3.3 V or the 5 V input, never both at the same time, and never apply 5 V to the 3.3 V input — doing so will damage the module. 3.3 V is the recommended supply. With either rail, decouple close to the module and provide transient and surge protection: supply ripple couples into the receiver, reducing effective range and increasing false triggers.

3.4 Launching the Application

1. Double-click KW007_DemoUpDn_v2.7.exe to start the application
2. The main window will appear with all controls

4. GUI Overview

The window is arranged in two halves. Everything a first-time user needs sits along the top;



the lower half stays disabled until "Distance Range" is set to (custom), and is there for fine-tuning.

*USB-to-I²C adapter: FT232H or CH341.

*Distance Range: Range1 is shortest, and Range8 is longest.

4.1 For “Quick Starters”

Three steps are enough to get the sensor running:

Step 1 — Click “Set Initial” to load the start-up configuration into the sensor.

Step 2 — Pick a “Distance Range”. Range1 is the shortest and Range8 the longest.

Step 3 — Turn Detection on. The two circle indicators light up as soon as the sensor reports movement.

4.2 For “Advanced Experts”

Set “Distance Range” to (custom) to enable the lower half of the window, then adjust the individual settings described in Section 5. You can also start from one of the eight ranges and fine-tune from there.

5. Custom Adjust Settings

Users can customize settings following the initial setup, or fine-tune them after configuring the distance range.

Detection Range

The detection range depends on two key parameters:

- Amplifier Gain: Increasing the gain extends the range.
- Magnitude Threshold: Lowering the threshold extends the range.

While higher gain or lower thresholds increase range, they also heighten susceptibility to false triggers.

Amplifier Gain Configuration

The total system gain is determined by summing the gain values of the three amplifier stages:

$$Total\ Gain = Amp_1\ Gain + Amp_2\ Gain + Amp_3\ Gain$$

Gain Selection Table

Bit Value	Amp ₁ Gain (dB)	Amp ₂ Gain (dB)	Amp ₃ Gain (dB)
000	8	8	0
001	12	12	4
010	16	16	8
011	19	19	12
100	21	21	-
101	24	24	-
110	28	28	-
111	32	32	-

For optimal Signal-to-Noise Ratio (SNR), it is recommended to apply higher gain settings in the earlier stages ($Amp_1 > Amp_2 > Amp_3$).

Amp3 Gain is a two-bit field: use 00, 01, 10 and 11 for 0, 4, 8 and 12 dB.

Magnitude Threshold Configuration

A target is detected only when the signal magnitude exceeds this threshold. For optimal results, follow the recommended setup sequence below:

1. Measure Baseline RSSI

The “Received Signal Strength Indicator” (RSSI) is the strength of the reflected signal: it rests at the background level when no target is present and rises when one moves within range. The RSSI chart in the lower-right panel shows it live.

2. Set “Mag Threshold” Above Noise Floor

3. Refer to the live RSSI to tune “Mag Threshold”: set it above the noise floor, but below the level reached when a target moves at the farthest distance that must still be detected. Changing the amplifier gain moves both readings, so re-check the threshold afterwards.

Note:

- Signal Saturation:

Improper gain causes signal saturation (clipping at 0 or 255), leading to detection failure.

- False Triggers:

If “Mag Threshold” is too close to baseline RSSI, environmental noise will trigger false detections.

- “Enhanced Range Mode” uses optimized parameters to boost total detection distance.

Doppler Filter Corner

This section configures the corner frequencies for the Doppler filters:

Filter Type	Function	Trade-off / Impact
Low-Pass Filter (LPF)	Attenuate high-speed clutter and high-frequency noise	Setting too low may attenuate fast-walking targets
High-Pass Filter (HPF)	Reject slow-motion background clutter	Setting too high may miss subtle, slow human movements

Sampling Rate

Key considerations for sampling rate adjustment:

- Filter Scaling:
The Doppler filter corner frequencies scale proportionally with the sampling rate.
- Noise Reduction:
A higher sampling rate mitigates noise aliasing, effectively lowering the in-band background noise floor.

GPO Delay

Specifies the hold time for GPO.

- Idle(GPO low): Default state when no motion is detected.
- Trigger(GPO active high): GPO is triggered instantly upon motion detection.
- Release: GPO is held active after motion stops, returning to Idle after delay expires.

Detect Period

I²C polling interval, adjustable from 10 to 1000 ms. The default of 128 ms matches the sensor's update rate; polling faster returns unchanged values.

Save / Load

Save writes the current Custom Adjust Settings and the Detect Period to an .ini file. Load reads one back, applies it to the sensor immediately, and reports the result in the message area. Use this to carry a tuned configuration between installations.

6. Monitoring

Approach and Leaving indicators:

Indicator	Color	Status
Upper	Green	Target Approaching
Lower	Blue	Target Leaving
Both	Grey	Idle (No Detection)

Signal timer:

The timer accumulates only while either indicator is lit, so it shows the total time the sensor has reported motion rather than the elapsed run time. Reset returns it to zero.

RSSI chart:

The chart plots reflection strength over the most recent 100 samples, oldest on the left. The vertical scale is 0 to 255; the horizontal scale is in samples, 20 per division. The number shown as “Value:” is the latest reading.

IIR low pass filter:

Signal Smoothing: Lower coefficient values apply stronger smoothing, while higher values track raw data closer.

7. Message Area

The message area at the lower left reports the outcome of every operation: which adapter was detected, whether Set Initial and each setting change were written and verified, and whether the sensor has been disconnected. Check it after any action that does not visibly change the window.

8. Troubleshooting

Message	Cause / Action
"FT232H not found" / "CH341 not found"	The adapter is not plugged in, or Windows has not enumerated it. Reconnect it and try again.
"... busy (used by another app?)"	Another program already has the adapter open. Close it and try again.
"driver missing (ftd2xx.dll)" / "driver missing (CH341DLL.dll)"	Install the FTDI D2XX driver for the FT232H, or the WCH CH341PAR driver for the CH341.
"... disconnected", followed by "stopped, controls unlocked"	The adapter was removed while running. Detection stops and the controls are released.
"Set Initial => FAIL", or a line such as "REG27 FAIL: w=0x30 r=0x00"	The value written did not read back. Check the wiring and the supply, then retry.
Settings appear to be accepted but the sensor never reports motion	Check that the module's Range1 pin is tied to GND. Without it the sensor runs in hardware mode and ignores the configuration.